Least-Cost Wildlife Corridor Modeling in Centre County, Pennsylvania: Assessing the Impact of Roads and Development on Habitat Connectivity

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Abstract

Habitat fragmentation is a major challenge for wildlife in developed regions, where roads, urban areas, and agriculture divide natural landscapes into isolated patches. This project aims to model potential wildlife corridors between Rothrock State Forest and Bald Eagle State Forest in Centre County, Pennsylvania using a least-cost corridor approach in ArcGIS Pro. Five spatial datasets were used to construct a resistance surface representing the relative difficulty of wildlife movement across the landscape: land cover (NLCD 2021), protected areas (PAD-US), road networks (TIGER/Line), elevation (USGS 3DEP), and population density (GPWv4). Each dataset was reclassified to a common 1–10 resistance scale and combined into a weighted composite resistance surface using the Raster Calculator tool. Two modeling scenarios were developed and compared: Scenario 1 representing an ideal landscape without direct road barriers, and Scenario 2 incorporating road network resistance. Cost Distance and Corridor tools were applied to each scenario to identify least-cost movement pathways between the two state forests. Results indicate that the primary corridor follows forested ridgelines along Bald Eagle Mountain, avoiding the developed Nittany Valley. The comparison of scenarios revealed that while overall corridor placement remained largely consistent, road barriers reduced corridor suitability around the State College urban area, identifying this zone as the primary pinch point where wildlife movement is most constrained. A subsequent fragmentation analysis physically delineated corridor segments separated by road barriers, confirming that Interstate 99, Route 322, and Route 45 collectively fragment the optimal corridor into isolated patches with limited opportunities for safe wildlife crossing near State College. These findings suggest that targeted conservation interventions such as wildlife crossings in the State College area could significantly improve landscape connectivity between the two core habitat areas.

Introduction

Habitat fragmentation is one of the most significant threats to biodiversity in developed landscapes. As roads, urban expansion, and agricultural land use divide once-continuous natural areas into isolated patches, wildlife populations become separated from one another, reducing genetic diversity, limiting access to resources, and increasing vulnerability to local extinction events (Xu et al., 2024). In regions like central Pennsylvania, where a mosaic of forested ridges, agricultural valleys, and growing urban centers defines the landscape, the challenge of maintaining ecological connectivity between core habitat areas is particularly pressing. Pennsylvania sits within the broader Appalachian corridor, one of the most ecologically significant wildlife movement routes in the eastern United States, making the preservation of local connectivity in counties like Centre especially important for regional biodiversity.

Wildlife corridors are connected strips or zones of suitable habitat that allow animals to move between larger isolated habitat patches and represent one of the most widely applied conservation tools for addressing the fragmentation described above. By identifying and protecting pathways through which wildlife can safely travel, corridor conservation efforts can help maintain population connectivity (Beltrão et al., 2024). Without functional corridors, isolated wildlife populations face increased risk of inbreeding, reduced resilience to environmental disturbances, and eventual local extinction. Roads are among the most significant barriers to corridor function, acting not only as physical obstacles but also as sources of noise, light, and chemical pollution that discourage wildlife from attempting to cross even where gaps in traffic exist.

This project focuses on Centre County, Pennsylvania, where Rothrock State Forest and Bald Eagle State Forest represent two large core habitat areas separated by the developed Nittany Valley and a dense transportation network. Rothrock State Forest encompasses approximately 95,000 acres of mixed hardwood forest in the ridge and valley landscape of central Pennsylvania (DCNR, 2024), providing habitat for a diverse assemblage of wildlife including white-tailed deer, black bear, wild turkey, and bobcat. Bald Eagle State Forest, covering approximately 193,000 acres to the northeast, represents

another significant block of contiguous forest habitat in the region (DCNR, 2024). Despite their size and ecological significance, these two forests are separated by the urbanized Nittany Valley, home to the State College metropolitan area and a dense network of roads including Interstate 99 and US Route 322. Least-cost corridor modeling is a GIS-based approach that identifies the most efficient movement pathway between two habitat areas by calculating the cumulative resistance cost of traveling across a landscape surface. Using this approach, this study aims to identify the most suitable wildlife movement pathways between these two forests and evaluate how roads and development influence corridor placement. Specifically, this study addresses the following research question: How do road networks and urban development influence the location, width, and continuity of least-cost wildlife corridors between Rothrock State Forest and Bald Eagle State Forest in Centre County, Pennsylvania?

The conservation of wildlife corridors in Pennsylvania is particularly timely given the ongoing development pressures facing the state. Centre County has experienced steady population growth driven by the expansion of Penn State University and associated commercial and residential development in the State College metropolitan area. This growth has accelerated the conversion of agricultural and semi-natural land cover in the Nittany Valley into developed land uses, reducing the permeability of the landscape for wildlife movement and increasing the isolation of forested habitat patches on the surrounding ridgelines. At the same time, the ridge and valley landscape of central Pennsylvania retains extensive areas of contiguous forest cover on its mountainous ridges, providing a foundation of relatively intact habitat that could support functional wildlife corridors if movement pathways through the developed valleys can be identified and protected. This project aims to contribute spatial evidence to support that goal by identifying where corridor connectivity is strongest, where it is most constrained, and where conservation investments would have the greatest impact.

Previous Research

Wildlife corridor modeling has been an active area of research in conservation biology and landscape ecology for several decades, with GIS-based least-cost path approaches becoming the dominant methodology for identifying potential movement pathways across complex landscapes. The foundational premise is that wildlife movement follows paths of least resistance, avoiding areas of high human disturbance, inhospitable land cover, and physical barriers such as roads and urban development (Liu et al., 2018).

Several key studies inform the methodological and conceptual foundation of this project. Liu et al. (2018) demonstrated the application of least-cost corridor modeling for restoring regional habitat connectivity, showing that the selection of resistance values and the weighting of individual landscape factors can significantly influence the location and characteristics of identified corridors. Their work highlighted the importance of using ecologically meaningful resistance values grounded in species-specific habitat requirements and emphasized that corridor models should be validated against known animal movement data where possible. The findings of Liu et al. are directly reflected in this study's approach of assigning ecologically grounded resistance values to each land cover class and protected area category rather than using arbitrary numerical assignments. Their emphasis on model validation is also echoed in this project's use of two scenario comparisons and a fragmentation analysis to cross check and strengthen the corridor modeling results.

Parks et al. (2013) examined how different weighting schemes applied to resistance surface variables affect corridor identification outcomes, finding that the relative importance assigned to land cover, roads, and terrain can produce substantially different corridor pathways. Their findings underscore the sensitivity of least-cost models to methodological decisions and suggest that comparing multiple modeling scenarios is an important step in understanding the uncertainty inherent in corridor predictions. This study directly applied the lesson from Parks et al. by developing two distinct modeling scenarios rather than relying on a single corridor output, allowing the influence of road barriers on corridor placement and suitability to be

explicitly quantified and mapped. The weighting scheme used in this study, assigning land cover the highest weight followed by slope and protected area status, was informed by the same ecological reasoning that Parks et al. identified as the most defensible basis for resistance surface construction.

Beltrão et al. (2024) contributed to the literature on priority area identification for ecological corridor implementation, demonstrating that spatially explicit corridor models can effectively guide conservation investments by identifying the locations where habitat connectivity is most critical and most threatened, directly informing the pinch point identification approach used in this study. Their work showed that corridor models are most useful for conservation planning when they move beyond simply identifying suitable habitat to explicitly locating bottlenecks and barriers that most severely constrain wildlife movement. This finding motivated the fragmentation analysis conducted in the present study, which went beyond standard corridor suitability mapping to physically delineate the discrete habitat fragments separated by road barriers and identify the specific road segments most responsible for severing corridor connectivity in Centre County.

Xu et al. (2024) examined the effects of habitat fragmentation on habitat quality, finding that fragmentation not only reduces the total area of suitable habitat but also degrades the quality of remaining patches through edge effects and isolation, reinforcing the conservation significance of maintaining connectivity between Rothrock and Bald Eagle State Forests. Their research provides important context for interpreting the fragmentation results of the present study, as the two isolated corridor fragments identified near State College are not simply reduced in area but are likely also degraded in ecological quality by their isolation from one another and from the larger forest blocks at each end of the corridor. This suggests that the conservation urgency of addressing the State College pinch point extends beyond restoring physical connectivity to also reversing the ecological degradation that prolonged isolation of these fragments may be causing for wildlife populations attempting to use them as movement habitat.

Methodology

Data

This project utilized five spatial datasets to model wildlife corridor suitability across Centre County, Pennsylvania. All datasets were sourced through the Esri Living Atlas of the World, accessible directly within ArcGIS Pro, after initial attempts to download datasets from their primary sources proved unsuccessful due to projection and rendering issues. The study area is shown in Figure 1.

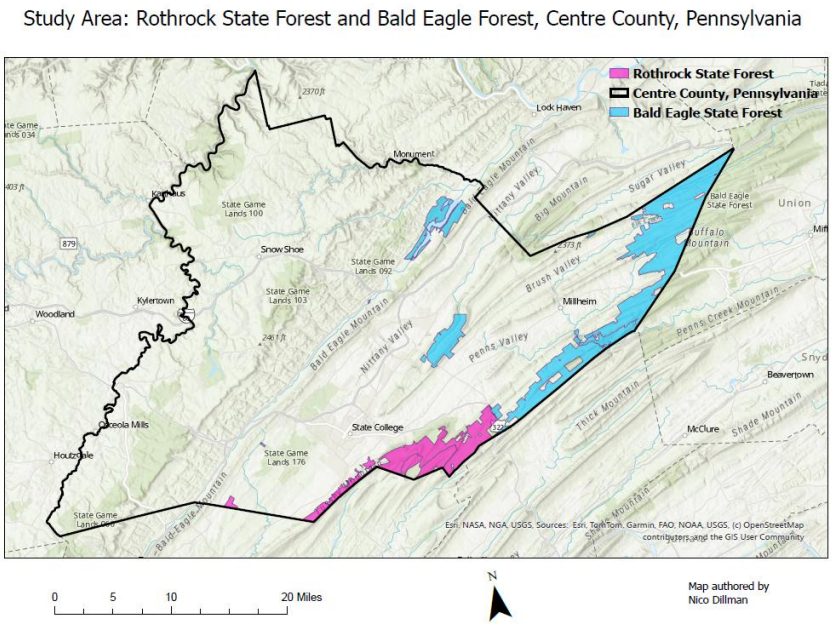


Figure 1: Study area map showing Rothrock State Forest (pink) and Bald Eagle State Forest (light blue) as core habitat areas within Centre County, Pennsylvania (black boundary).

Land cover data was obtained from the USA Annual NLCD Land Cover layer (MRLC, 2021), which classifies land surface into 16 categories including various forest types, developed intensities, agricultural land, wetlands, and open water. Land cover was selected as the primary resistance variable as it directly reflects habitat quality and human disturbance intensity across the landscape, making it the most ecologically meaningful input for corridor modeling. Protected areas data was sourced from the PAD-US Protection Status by GAP Status Code layer (USGS, 2024), which classifies protected lands into four GAP status categories ranging from strictly protected biodiversity areas to unprotected lands. Protected areas were included because they represent low-resistance, high-quality habitat where wildlife movement is least constrained by human activity. Road network data was obtained from the TIGER/Line Roads shapefile (U.S. Census Bureau, 2025) for Centre County, representing the transportation network as a barrier to wildlife movement. Roads were included as a separate resistance variable because they represent one of the most significant physical barriers to wildlife movement in developed landscapes. Elevation data was sourced from the USGS 3DEP Elevation layer at 30-meter resolution, providing terrain information across the study area. Elevation and slope were included as resistance variables because steep terrain presents physical movement challenges for most wildlife species. Population density data was obtained from the Gridded Population of the World Version 4 (GPWv4) dataset (NASA SEDAC, 2020) at 1km resolution, representing human disturbance as a proxy variable. Population density was included in the original proposal as an indicator of human activity levels across the landscape, with the expectation that areas of higher population density would correspond to greater resistance to wildlife movement.

Several data preparation challenges were encountered during this phase. The NLCD GeoTIFF downloaded directly from the MRLC Viewer failed to render correctly in ArcGIS Pro despite previewing accurately on the website, displaying only a single land cover class over a small portion of the county boundary. Similarly, the PAD-US geodatabase downloaded from the USGS produced broken layer file references that could not be resolved. As a result, all primary datasets were ultimately sourced through the Esri Living Atlas, which provided correctly projected and immediately usable layers. The roads shapefile required reprojection from GCS_North_American_1983 to the USA Contiguous Albers Equal Area Conic

coordinate system using the Project tool prior to further processing, as the original projection conflicted with the other layers and produced empty outputs during raster conversion. The GPWv4 population density layer, while successfully downloaded and exported as a local raster at 1km resolution, ultimately could not be incorporated into the final resistance surface due to its coarse spatial resolution relative to the 30-meter resolution of the other datasets, which generated widespread NoData gaps when combined in the overlay analysis. This limitation is discussed further in the Reflection section.

Although population density was originally included as a key resistance variable in the project proposal, it was ultimately excluded from the final analysis due to its coarse spatial resolution (1 km), which was incompatible with the 30-meter resolution of the other datasets. This mismatch produced extensive NoData gaps during raster overlay operations, preventing successful integration. As a result, the final resistance surface reflects a modified implementation of the proposed methodology. All datasets represent a single time-period (primarily around 2020 to 2021), so the analysis reflects current landscape conditions rather than changes over time.

Methods

The analytical workflow followed a multi-step process to develop a resistance surface representing the relative difficulty of wildlife movement across Centre County, culminating in a least-cost corridor model identifying potential movement pathways between Rothrock State Forest and Bald Eagle State Forest.

Data Preparation and Reclassification

Prior to analysis, all vector layers were converted to raster format to enable raster-based calculations. The PAD-US protected areas layer was converted using the Polygon to Raster tool in ArcGIS Pro, with the GAP Status Code field selected as the value field and a cell size of 30 meters to maintain consistency with the NLCD resolution. The roads shapefile was converted using the Polyline to Raster tool with the RTTYP field, also at 30-meter resolution. Following conversion, each raster dataset was reclassified to a common 1–10 resistance scale using the Reclassify tool, where values of 1 represent conditions of minimal resistance to wildlife movement and values of 10 represent maximum resistance or complete barriers.

Land cover classes were reclassified based on their relative suitability for wildlife movement (Table 1). Deciduous and evergreen forest received the lowest resistance values of 1, reflecting their high habitat suitability, while mixed forest and wetland classes received a value of 2. Shrub and scrub received a value of 3, grassland and herbaceous cover a value of 4, and developed open space and pasture a value of 5. Barren land and cultivated crops were assigned a value of 6, developed low intensity a value of 7, open water a value of 8 to reflect its barrier function for terrestrial wildlife, developed medium intensity a value of 9, and developed high intensity the maximum value of 10.

Table 1: Land Cover Reclassification Values

Land Cover Class	Original NLCD Code	Resistance Value
Deciduous Forest	41	1
Evergreen Forest	42	1
Mixed Forest	43	2
Woody Wetlands	90	2
Emergent Herbaceous Wetlands	95	2
Shrub/Scrub	52	3
Grassland/Herbaceous	71	4
Developed, Open Space	21	5
Pasture/Hay	81	5
Barren Land	31	6
Cultivated Crops	82	6
Developed, Low Intensity	22	7
Open Water	11	8
Developed, Medium Intensity	23	9
Developed, High Intensity	24	10

Protected area classes were reclassified according to their GAP status codes (Table 2). Strictly protected areas received a resistance value of 1, managed protected areas a value of 3, multiple use areas a value of 6, and unprotected lands a value of 8, reflecting the assumption that protection status correlates positively with habitat quality and wildlife movement suitability.

Table 2: PAD-US GAP Status Code Reclassification Values

GAP Status Code	Description	Resistance Value
1	Strictly Protected – Biodiversity Focus	1
2	Managed Protected – Natural Disturbance Suppressed	3
3	Multiple Use – Extraction Permitted	6
4	Unprotected	8

Road classes were simplified from six TIGER/Line Road type categories into three barrier classes. Interstate and major roads were merged into a high barrier class with a resistance value of 10, secondary and other roads into a medium barrier class with a value of 6, and unknown and local roads into a low barrier class with a value of 3. For elevation, the Slope tool was first applied to derive terrain steepness in degrees from the raw DEM, which was then reclassified into six categories as shown in Table 3.

Table 3: Slope Reclassification Values

Slope Range (Degrees)	Description	Resistance Value
0 – 5	Flat	1
5 -15	Gentle Slope	3
15 – 25	Moderate Slope	5
25 – 35	Steep Slope	7
35 – 45	Very Steep	9
45 – 90	Near Vertical	10

Resistance Surface Construction

Following reclassification, the individual resistance layers were combined into a single composite resistance surface using the Raster Calculator tool in ArcGIS Pro. The Weighted Overlay tool was initially attempted but produced incomplete outputs due to NoData conflicts between layers, particularly from the roads raster which contained valid values only where road pixels existed and generated NoData in all other areas, causing widespread data gaps across the county. The Raster Calculator provided a more flexible alternative, allowing explicit handling of NoData areas through conditional statements.

The baseline resistance surface (Scenario 1 — No Roads) was calculated using the following conditional expression:

Con (IsNull("Reclass_PAD_Centre_County"), ("Reclass_Land_Cover" x 0.70) + ("Reclass_Slope" x 0.30), ("Reclass_Land_Cover" x 0.45) + ("Reclass_PAD_Centre_County" x 0.25) + ("Reclass_Slope" x 0.30))

This expression uses a conditional (Con) statement to handle NoData gaps in the PAD-US layer. Where the PAD-US raster contained NoData, the formula applied a two-variable weighted combination of land cover (70%) and slope (30%) to ensure complete spatial coverage. Where PAD-US data was present, the full three-variable weighted formula was applied, assigning land cover a weight of 45%, PAD-US a weight of 25%, and slope a weight of 30%. Land cover received the highest weight given its direct representation of habitat quality and movement suitability.

The decision to assign land cover the highest weight of 45% reflects its status as the most direct and ecologically meaningful indicator of habitat quality and movement suitability available in this analysis. Land cover directly captures the physical character of the landscape surface that wildlife must navigate, distinguishing between forested areas that provide cover and resources, agricultural areas that offer limited but not negligible permeability, and developed areas that present near-complete barriers to movement. The PAD-US protection status layer was assigned a weight of 25% to reflect its importance as a secondary modifier of habitat quality, recognizing that protected lands tend to support higher quality wildlife habitat due to reduced human disturbance and more intact ecological conditions. Slope was assigned a weight of 30% as a terrain-based resistance variable, reflecting the energetic cost that steep terrain imposes on wildlife movement regardless of land cover type. A flat agricultural field and a steep forested slope may have similar land cover resistance values, but the steep terrain imposes an additional physical cost that the slope weight accounts for. The 50-meter buffer applied to roads during the fragmentation analysis was chosen to represent the zone of direct physical and perceptual disturbance immediately adjacent to road surfaces, within which wildlife movement is most strongly deterred by traffic noise, vehicle lights, and the physical barrier of the road surface itself. While the optimal buffer distance varies by species and road type, 50 meters represents a conservative and commonly applied threshold in landscape connectivity analyses. The resulting Scenario 1 resistance surface contains values ranging from 1 to 9.

To create the Scenario 2 (With Roads) resistance surface, the roads raster was incorporated into the baseline surface using the following Raster Calculator expression:

Con (IsNull("Roads_Project_PolylineToRaster"), "Resistance_Surface_Scenario1", ("Resistance_Surface_Scenario1" + 3))

This expression adds a resistance value of 3 to all pixels where roads exist, pushing road pixels to a maximum resistance of approximately 11.5 while leaving all non-road pixels unchanged. The resulting Scenario 2 resistance surface contains values ranging from 1 to 11.5.

Corridor Modeling

Least-cost corridor modeling was performed for both scenarios using the Cost Distance and Corridor tools in ArcGIS Pro. Core habitat source and destination areas were defined by extracting the Rothrock State Forest and Bald Eagle State Forest polygons from the PAD-US layer using Select by Attributes and Copy Features tools. The Cost Distance tool was applied twice for each scenario, once using Rothrock State Forest as the source and once using Bald Eagle State Forest as the source, producing four cumulative cost distance surfaces in total (Figures 5 and 6). The Corridor tool was then applied to each pair of cost distance surfaces, adding them together to produce a corridor suitability surface for each scenario where the lowest summed values represent the optimal movement zone between the two forests (Figures 7 and 8). Finally, a spatial comparison of the two corridor outputs was conducted using the Raster Calculator to subtract the Scenario 1 corridor from the Scenario 2 corridor, producing a difference raster highlighting areas of greatest divergence between the two modeled pathways (Figure 9). Results were communicated through a series of static maps, including resistance surfaces, cost distance outputs, corridor maps, and comparison maps.

Corridor Fragmentation Analysis

To physically delineate road-induced corridor breaks, the optimal corridor zone was extracted from the Scenario 1 corridor raster and converted to a polygon using the Raster to Polygon tool. Roads were buffered by 50 meters using the Buffer tool and subsequently erased from the optimal corridor polygon using the Erase tool, producing discrete corridor fragments representing contiguous habitat patches separated by road barriers.

The 50-meter buffer distance applied to roads was selected based on general principles of road ecology and wildlife movement behavior. Research on wildlife responses to roads consistently shows that the barrier effect of a road extends beyond the physical road surface itself, with traffic noise, artificial lighting, and vehicle emissions creating a zone of disturbance that deters wildlife movement in the immediate vicinity of the road even where no physical barrier exists. A 50-meter buffer was chosen as a conservative estimate of this disturbance zone, capturing the area immediately adjacent to road surfaces where wildlife movement is most directly suppressed by road-related disturbance without overstating the spatial extent of road impacts across the broader landscape. For major highways such as Interstate 99, which carry significantly higher traffic volumes and generate greater levels of noise and disturbance than local roads, a larger buffer of 100 meters or more might be more ecologically appropriate and could be explored in future analyses. However, applying a uniform 50-meter buffer across all road classes provided a consistent and reproducible methodological approach that avoided introducing additional uncertainty from assumptions about road-specific disturbance distances. The resulting buffered road polygon effectively captured the zone of direct road influence across Centre County, producing corridor fragments that reflect not just the physical road surface, but the broader zone of movement suppression associated with road infrastructure.

RStudio Analysis

To complement the spatial analysis conducted in ArcGIS Pro, summary visualizations were produced in RStudio using the ggplot2 package to graphically illustrate the resistance value assignments underlying the corridor model and to support the interpretation of corridor modeling results in a format accessible to non-GIS audiences. A horizontal bar chart was created displaying the resistance values assigned to each NLCD land cover class, providing a clear visual representation of the resistance hierarchy applied across land cover types. A pie chart was also produced summarizing the distribution of the 15 NLCD land cover classes across three broad resistance categories: low resistance representing values 1 to 3, moderate resistance representing values 4 to 6, and high resistance representing values 7 to 10. Finally, a grouped bar chart was produced comparing the relative area of each corridor suitability zone between Scenario 1 and Scenario 2, providing a quantitative visual summary of how road barriers influence the distribution of corridor suitability across the study area. All three visualizations were produced to support the interpretation of the resistance surface methodology and corridor modeling results.

Results

Resistance Surface

The resistance surface produced for both scenarios effectively captured the spatial variation in landscape permeability across Centre County, Pennsylvania (Figure 2). Under Scenario 1 (No Roads), resistance values ranged from 1 to 9, with the lowest values concentrated along the forested ridgelines of Rothrock State Forest in the western portion of the county and along Bald Eagle Mountain in the northeast (Figure 2, left panel). These areas of low resistance reflect the dominance of deciduous and mixed forest cover combined with protected land status, which together create highly permeable habitat for wildlife movement. The Nittany Valley corridor running diagonally through the center of the county displayed moderate to high resistance values, reflecting the dominance of agricultural land use, developed open space, and scattered residential development. The State College urban area produced the highest resistance values on the surface, reflecting the concentration of high-intensity developed land cover in this portion of the county.

What is particularly striking about the State College area is that its high resistance is not solely a function of roads and buildings. The city is home to Penn State University, one of the largest universities in the United States, which drives a resident and student population of well over 100,000 people in and around

the borough. This level of human activity generates constant noise, light, foot traffic, and vehicle movement at all hours of the day and night, creating an environment that is fundamentally inhospitable to most wildlife species regardless of the physical infrastructure present. Unlike a rural road that wildlife might attempt to cross during low traffic periods at night, the State College urban core presents a near-continuous wall of human activity that effectively eliminates the possibility of safe wildlife movement through the area at any time of day. This is precisely why population density, had it been successfully incorporated into the resistance surface at an appropriate resolution, would likely have significantly amplified the resistance values in and around State College beyond what land cover alone captures.

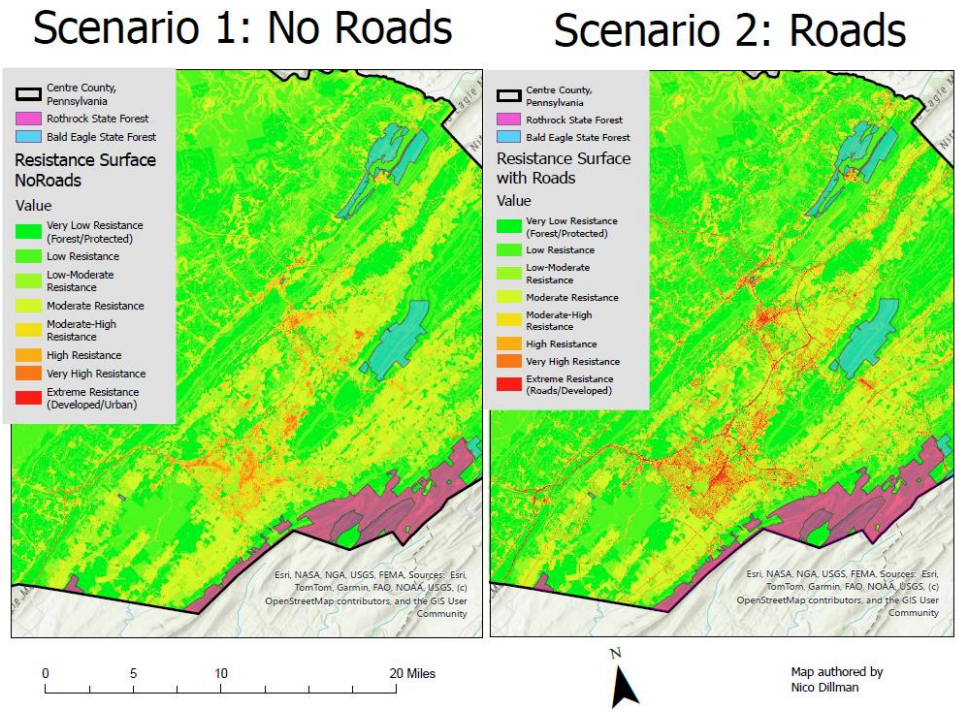


Figure 2: Resistance surfaces for Scenario 1 — No Road Barriers (left) and Scenario 2 — With Road Barriers (right). Values range from Very Low Resistance (dark green) to Extreme Resistance (red).

Under Scenario 2 (With Roads), the resistance surface retained the same overall spatial pattern but introduced additional resistance along the road network, pushing road pixel values above 9 and creating a new maximum resistance category ranging from 9 to 11.5 (Figure 2, right panel). The road network was most visible as a source of additional resistance in Nittany Valley where the density of primary and secondary roads is highest, and along Interstate 99 and US Route 322 which bisect the county. The addition of road resistance did not dramatically alter the overall pattern of the surface but introduced localized zones of increased barrier effect concentrated along major transportation corridors.

To further illustrate the resistance value hierarchy applied across the landscape, a horizontal bar chart was produced in RStudio summarizing the resistance values assigned to each NLCD land cover class (Figure 3). The chart clearly shows the full range of resistance values applied across the 15 land cover classes, with Deciduous Forest and Evergreen Forest anchoring the low end of the scale at a resistance value of 1 and Developed High Intensity occupying the maximum value of 10. Visualization reinforces the ecological logic underlying the resistance surface, showing a clear and consistent progression from natural and forested land cover types at the low resistance end to intensively developed and impervious land cover types at the high resistance end. Wetland classes cluster near the low end of the scale alongside forest types, reflecting their value as wildlife habitat, while agricultural classes such as cultivated crops and pasture fall in the moderate resistance range, reflecting their reduced but not negligible permeability for wildlife movement.

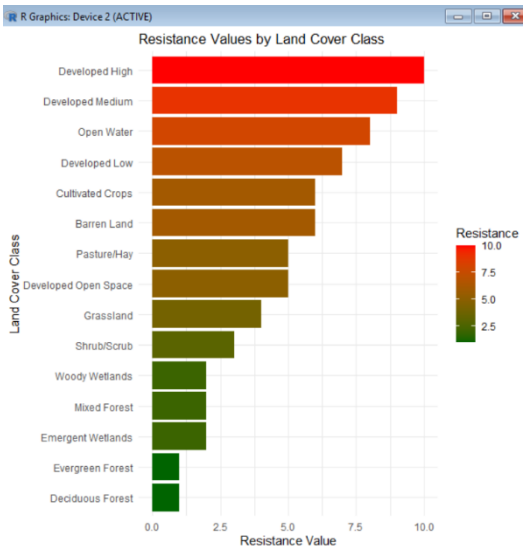


Figure 3: Resistance values assigned to each NLCD land cover class, ranging from Deciduous Forest (lowest resistance) to Developed High Intensity (highest resistance).

The distribution of land cover classes across three broad resistance categories is further summarized in the pie chart shown in Figure 4. Of the 15 NLCD land cover classes present in Centre County, six fall into the low resistance category representing values 1 to 3, five fall into the moderate resistance category representing values 4 to 6, and four fall into the high resistance category representing values 7 to 10. This roughly equal distribution across resistance tiers reflects the ecological diversity of Centre County, where extensive forested ridgelines and protected areas coexist alongside a mix of agricultural valleys and developed urban centers. The balance between low, moderate, and high resistance land cover types across the county creates a complex movement landscape where wildlife corridor pathways are strongly influenced by the spatial arrangement of these land cover classes relative to one another and to the core habitat areas at Rothrock and Bald Eagle State Forests.

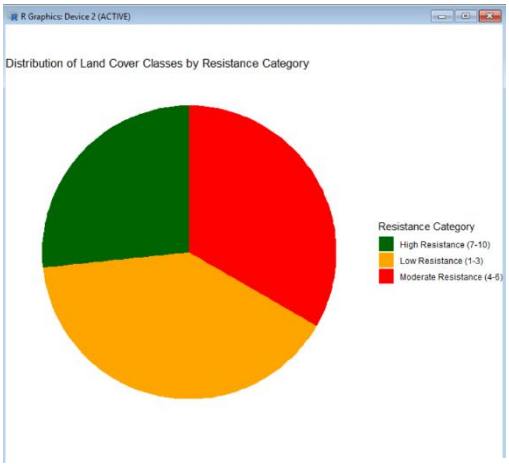


Figure 4: Distribution of NLCD land cover classes across low (1 to 3), moderate (4 to 6), and high (7 to 10) resistance categories.

Cost Distance Analysis

Two sets of cost distance surfaces were generated for each scenario to model the cumulative resistance cost of wildlife movement radiating outward from each core habitat area. Under Scenario 1 (No Roads), the cost distance surface originating from Rothrock State Forest produced cumulative values ranging from 0 within the forest boundary to a maximum of 146,990, while the surface originating from Bald Eagle State Forest ranged from 0 to 96,496 (Figures 5 and 6). Under Scenario 2 (With Roads), the Rothrock cost distance surface ranged from 0 to 112,327 and the Bald Eagle surface ranged from 0 to 71,267.

The spatial pattern of the cost distance surfaces reflects the underlying resistance surface and the ridge and valley landscape structure of Centre County. Areas of low cumulative cost, shown in green on the maps, are concentrated along the forested ridgelines immediately adjacent to each source forest, where continuous forest cover and protected land status combine to create highly permeable movement conditions. As distance from each source forest increases and the landscape transitions from forested ridgelines into the agricultural and developed Nittany Valley, cumulative cost values increase steadily,

reflecting the greater resistance encountered by wildlife attempting to move through these areas. The highest cumulative cost values, shown in pink and tan on the maps, are concentrated in the portions of the county most distant from each source forest and most dominated by developed land cover, particularly in the western portions of the county when viewed from Bald Eagle State Forest and in the northeastern portions when viewed from Rothrock State Forest.

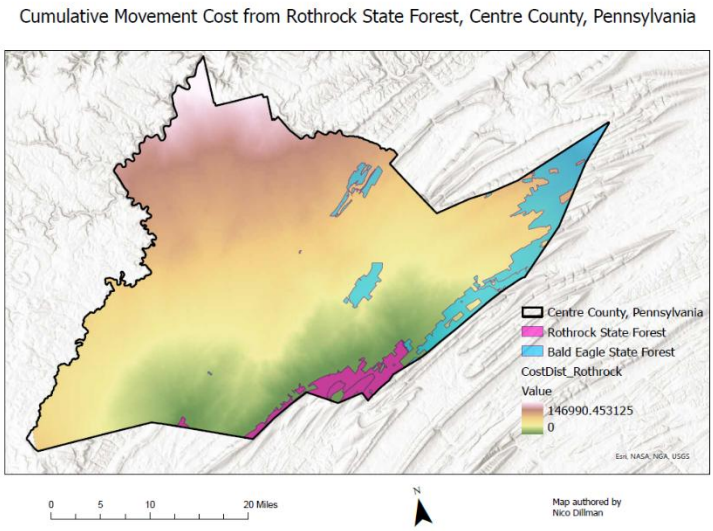


Figure 5: Cumulative least-cost movement surface from Rothrock State Forest. Green indicates low movement cost and pink/tan indicates high cumulative movement cost.

One of the most visually striking patterns in the cost distance maps is the clear contrast between the green low-cost zone surrounding Bald Eagle State Forest and the high-cost pink zone in the northwestern portion of the county when viewed from Bald Eagle (Figure 6). This pattern reveals that the landscape surrounding Bald Eagle State Forest is relatively accessible and well connected, with low cumulative movement costs extending across much of the eastern and central portions of the county along the forested ridgelines. However, once the cost surface crosses the State College urban area moving westward, cumulative costs rise sharply and the transition from green to yellow to pink happens rapidly, visually demonstrating how the concentration of development, roads, and human activity around State College acts as a wall that dramatically increases the cost of movement for any wildlife attempting to travel between the two forests. The constant activity generated by Penn State University and the surrounding student population, including vehicle traffic, pedestrian activity, noise, and artificial lighting at all hours, compounds the physical barrier effect of the road network and creates an environment where wildlife movement is effectively suppressed across a broad zone rather than just at individual road crossing points.

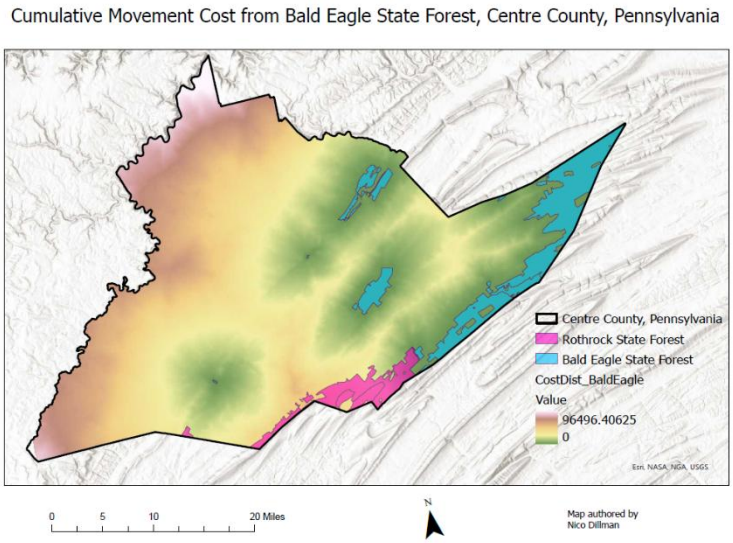


Figure 6: Cumulative least-cost movement surface from Bald Eagle State Forest. Green indicates low movement cost and pink/tan indicates high cumulative movement cost.

This same pattern is visible in the Rothrock cost distance surface (Figure 5), where the green low-cost zone radiates outward from Rothrock State Forest across the southern ridgelines and state game lands before encountering the sharply elevated costs of the Nittany Valley and State College area. The area north and northeast of State College, however, shows relatively lower cumulative costs compared to the area immediately around and south of the city, suggesting that the forested ridgeline corridor along Bald Eagle Mountain provides a viable lower cost movement pathway that bypasses the most developed portions of the valley. This finding reinforces the importance of the northern ridgeline corridor as the primary viable movement pathway between the two state forests and highlights why conservation efforts should focus on maintaining the integrity of this forested connection.

Notably, the maximum cumulative cost values were lower in Scenario 2 than in Scenario 1, which was an unexpected finding. Rather than uniformly increasing resistance across the landscape, the addition of road barriers appears to have redirected and concentrated modeled movement into fewer available pathways, effectively shortening the maximum cost distance while increasing resistance along those constrained routes. This result suggests that roads do not simply increase overall movement cost but fundamentally alter the spatial structure of movement opportunities across the landscape, channeling wildlife into narrower movement corridors between road barriers rather than allowing free movement across the full landscape. This finding is consistent with previous research showing that linear infrastructure such as roads can channelize wildlife movement into narrow corridor zones between barrier features (Liu et al., 2018) and highlights the importance of maintaining habitat quality in the remaining movement gaps between major road barriers in Centre County.

Wildlife Corridor Modeling

The least-cost corridor analysis identified a primary movement pathway between Rothrock State Forest and Bald Eagle State Forest running along the forested ridgelines of Bald Eagle Mountain and the associated state game lands in the northern and central portions of Centre County (Figures 7 and 8). Under both scenarios the corridor follows a broadly similar route, hugging the forested ridge systems and avoiding the developed Nittany Valley where resistance values are highest. The areas of optimal corridor suitability, represented by the lowest summed cost distance values, were concentrated along the ridge crests where forest cover is continuous and human disturbance is minimal. This result is consistent with

the underlying resistance surface, which assigned the lowest resistance values to forested and protected areas and the highest values to developed and agricultural land cover types.

The spatial alignment of the modeled corridor with the forested ridgelines of Bald Eagle Mountain and the associated state game lands is ecologically meaningful. These areas represent some of the largest and most contiguous blocks of forest habitat remaining in Centre County, providing not only low resistance movement conditions but also food resources, cover, and denning habitat for a wide range of wildlife species. The state game lands parcels distributed along the ridge systems further reinforce the corridor by providing legally protected land with minimal human disturbance, creating a network of stepping stone habitat patches that collectively support wildlife movement between the two core forest areas.

Under Scenario 1 (No Roads) the optimal corridor zone was slightly broader and more expansive, reflecting the absence of direct road barriers and representing an idealized movement landscape where wildlife can move freely across the terrain following the path of least resistance (Figure 7). The broader optimal zone under Scenario 1 suggests that in the absence of road barriers, wildlife would have access to a wider range of movement pathways through the landscape, increasing the redundancy and resilience of the corridor network. Redundancy in corridor networks is ecologically important because it reduces the vulnerability of wildlife populations to localized disturbances such as development, logging, or severe weather events that could temporarily block individual movement pathways.

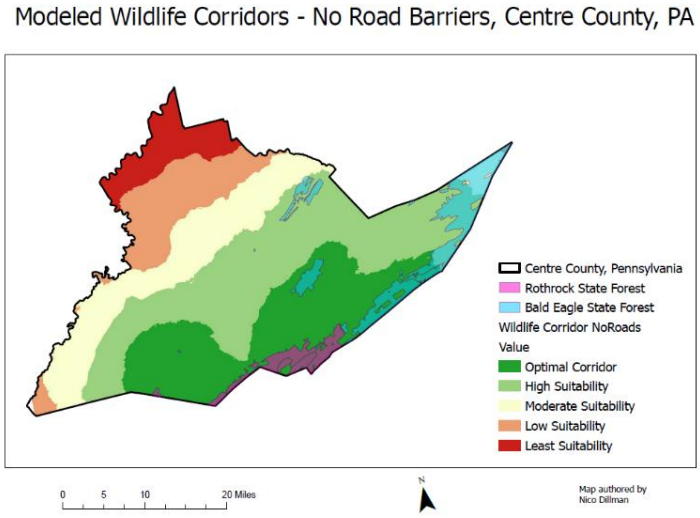


Figure 7: Modeled least-cost wildlife corridor under Scenario 1 — No Road Barriers, ranging from optimal suitability (dark green) to least suitable (red).

Under Scenario 2 (With Roads) the zones of highest suitability became slightly more constrained and narrowed in areas where the road network intersects potential movement pathways, reflecting the additional resistance imposed by roads and forcing modeled movement into fewer available gaps between road barriers (Figure 8). The narrowing of the optimal corridor zone under Scenario 2 is most pronounced in the central portion of the county where the road network is densest, particularly around the State College metropolitan area and along the Interstate 99 and Route 322 corridors. This narrowing effect reduces the functional width of the corridor available to wildlife, increasing the likelihood that animals attempting to cross between the two state forests will be forced into suboptimal movement conditions or deterred from attempting the crossing entirely.

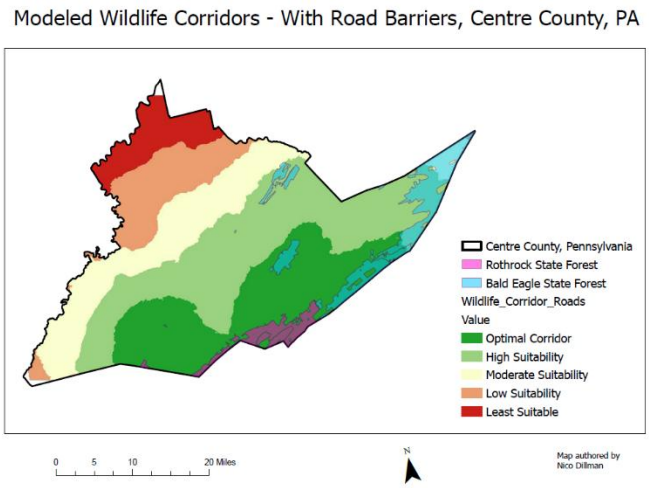


Figure 8: Modeled least-cost wildlife corridor under Scenario 2 — With Road Barriers, ranging from optimal suitability (dark green) to least suitable (red).

It is worth noting that when viewed independently, Figures 7 and 8 appear remarkably similar at first glance. The overall shape, location, and color pattern of the corridor looks nearly identical between the two scenarios, which might suggest that roads have little meaningful impact on corridor suitability. However, this visual similarity is somewhat misleading. When the two surfaces are overlaid and compared directly, as done in the corridor comparison analysis described in the following section, subtle but meaningful differences become apparent in the spatial extent and boundaries of each suitability zone. The optimal and high suitability zones are slightly smaller and more constrained under Scenario 2, while the moderate and low suitability zones expand slightly to fill the space where higher suitability areas are contracted. These differences are small enough to be nearly invisible when looking at each map independently but become clearly measurable when the two surfaces are subtracted from one another, highlighting the value of the scenario comparison approach for detecting road impacts that would otherwise be difficult to observe through visual map inspection alone.

Corridor Comparison and Pinch Point Identification

A spatial comparison of the two modeled corridor scenarios was conducted by subtracting the Scenario 1 corridor surface from the Scenario 2 corridor surface using the Raster Calculator tool, producing a difference raster that highlights areas of greatest divergence between the two modeled pathways (Figure 9). The resulting comparison map reveals that while the overall corridor placement between Rothrock and Bald Eagle State Forests remained largely consistent across both scenarios, meaningful spatial differences emerged in specific areas that align with the known landscape structure of Centre County. The grouped bar chart produced in RStudio further supports this finding by quantitatively illustrating how the distribution of corridor suitability zones shifted between the two scenarios (Figure 10). The chart shows that the optimal corridor zone covers a greater relative area under Scenario 1 compared to Scenario 2, while the

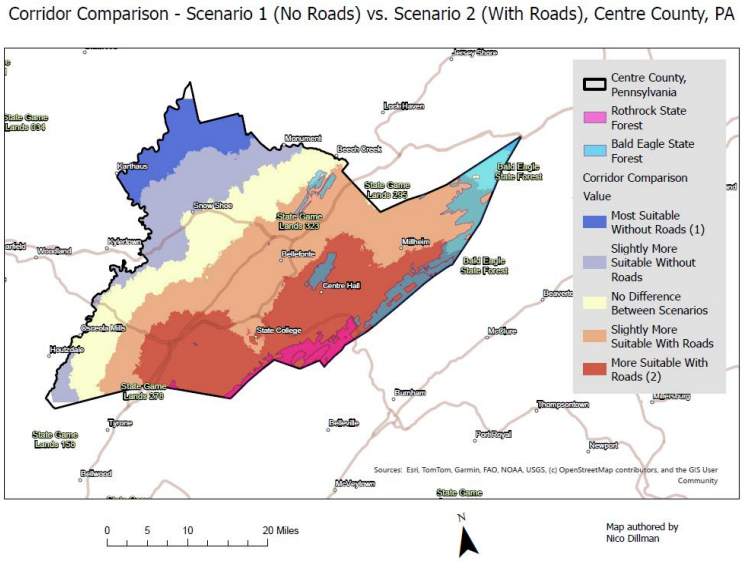


Figure 9: Corridor suitability comparison between Scenario 1 (blue) and Scenario 2 (red/brown), with yellow indicating no difference between scenarios.

least suitable zone covers a smaller relative area under Scenario 1, confirming that road barriers reduce the extent of high-quality corridor habitat across the study area.

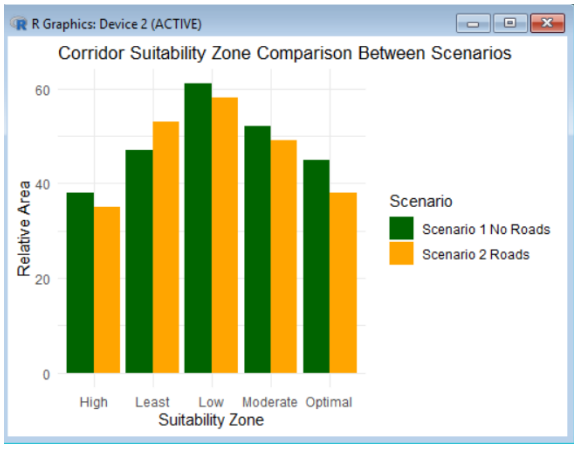


Figure 10: RStudio bar chart comparing corridor suitability zone areas between Scenario 1 (No Road Barriers) and Scenario 2 (With Road Barriers).

Looking at Figure 9 more closely, the spatial pattern of the comparison map tells a clear and intuitive story about the relationship between land cover, roads, and corridor suitability in Centre County. The deep blue area in the northwestern portion of the county, representing zones that are most suitable under Scenario 1 without roads, aligns almost perfectly with the most remote and heavily forested portions of the county where road density is lowest and human disturbance is minimal. These are the areas where the absence of roads makes the greatest difference to corridor suitability, because without road barriers wildlife can move freely through continuous forest cover with very little resistance. Moving southward and eastward toward Nittany Valley, the map transitions through lighter blue and yellow zones representing areas of slight advantage without roads and no meaningful difference between scenarios, before shifting into the orange and red zones that dominate the central and southern portions of the county.

The predominance of orange and red in the central and southern portions of the map, representing areas that score slightly better under Scenario 2 with roads, is initially counterintuitive but reflects the mathematical nature of the corridor calculation. When roads redirect modeled movement into fewer available pathways, the cost surface concentrates lower values along those constrained routes, which can actually improve relative corridor scores in specific locations even as overall corridor quality decreases. This is consistent with the unexpected finding from the cost distance analysis that maximum cumulative cost values were lower under Scenario 2 than Scenario 1.

When zoomed into the State College area, this effect is clearly visible as narrow lighter colored gaps threading through the dense red zone, representing the few remaining movement pathways the model identified between road barriers rather than genuinely high-quality habitat. The University Park Airport, Penn State Golf Courses, and the dense local road network including N Atherton Street and S Atherton Street further compound the barrier effect, collectively creating one of the most movement-hostile urban environments possible for wildlife attempting to cross between the two state forests.

The most notable area of divergence was identified in and around the State College urban core, where the addition of road barriers in Scenario 2 produced a measurable reduction in corridor suitability compared to the ideal landscape of Scenario 1. These results identify the State College metropolitan area as the primary pinch point in the modeled corridor network, a location where the road network and associated urban development most significantly constrain potential wildlife movement between the two state forests. State College presents a uniquely challenging barrier in this landscape due to its combination of high population density, a dense road network, and ongoing urban and suburban development pressure driven in part by the presence of Penn State University. Unlike rural road barriers that may consist of a single highway crossing, the State College urban core represents a broad zone of compounding resistance where high-intensity land cover, traffic volume, and infrastructure density combine to create a functionally impermeable barrier for wildlife attempting to move between the two state forests. As the area continues to grow, these barriers are likely to intensify, making early conservation intervention particularly urgent. While this result may appear intuitive given the density of roads and development in this area, the spatial analysis provides quantitative cartographic evidence of exactly where these

constraints occur and how significantly they alter modeled movement pathways, information that would not be available through visual inspection of the landscape alone.

In contrast, the northern portions of Centre County, characterized by extensive forested ridgelines, state game lands, and lower road density, showed little meaningful difference between the two scenarios. The results show that the forested northern landscape provides robust connectivity opportunities for wildlife regardless of road barrier effects, and suggests these areas represent priority zones for conservation and habitat protection efforts aimed at maintaining long-term landscape connectivity. The consistency of corridor suitability in the northern portion of the county across both scenarios is an encouraging finding, as it suggests that the existing network of state game lands and forested ridgelines in this area is sufficiently intact to support wildlife movement even in the presence of road barriers. Conservation efforts in Centre County should therefore prioritize both protecting these northern forest blocks from future development and addressing the road barrier effects concentrated in the State College area.

What Figure 9 ultimately reinforces is a finding that is both spatially explicit and ecologically intuitive. The northern forested portions of Centre County are where wildlife have the best chance of moving between the two state forests, and the State College urban area is where that movement breaks down most severely. All the data across every analytical step consistently points toward the same conclusion. As long as wildlife can navigate around or avoid the Penn State University area and the dense road network surrounding State College, viable movement pathways exist along the northern ridgelines connecting Rothrock and Bald Eagle State Forests. The challenge for conservation planning is therefore not identifying where the corridor should go, as the data makes that clear, but rather figuring out how to help wildlife get past the State College pinch point safely, whether through wildlife crossings, habitat restoration, or targeted land use planning along the key road corridors of Interstate 99, Route 322, and Route 45.

Overall, the results show that while roads do not fundamentally redirect the overall corridor pathway between Rothrock and Bald Eagle State Forests, they do meaningfully constrain corridor width and suitability in developed areas, particularly around State College. These findings support the targeted placement of wildlife crossing infrastructure or habitat connectivity initiatives at the identified pinch point locations, where relatively modest conservation investments could yield significant improvements in landscape permeability for wildlife moving between the two core habitat areas.

Corridor Fragmentation Analysis

To further examine the spatial impact of road barriers on wildlife movement, a fragmentation analysis was conducted on the optimal corridor zone identified under Scenario 1. Roads were buffered by 50 meters and erased from the optimal corridor polygon using the Erase tool in ArcGIS Pro, producing corridor fragments representing contiguous habitat patches separated by road barriers (Figures 11, 12, and 13). The fragmentation analysis builds directly on the corridor comparison results by moving beyond a relative comparison of suitability zones to physically delineating the discrete habitat patches that remain accessible to wildlife within the optimal corridor after road barriers are accounted for.

The analysis revealed that the optimal corridor is physically severed into two primary fragments by the road network crossing the Nittany Valley (Figure 11). The western fragment, located near State Game Lands 176 west of State College, is isolated from the eastern fragment by Interstate 99 and Route 322. This western fragment encompasses a substantial area of agricultural and semi-natural land cover west of the State College urban area, and while it represents suitable movement habitat in isolation, its disconnection from the eastern fragment means that wildlife moving through this area have no continuous pathway to reach Bald Eagle State Forest without crossing a major road barrier. The presence of State Game Lands 176 within this western fragment provides some degree of protected habitat, but the surrounding agricultural and developed land cover limits the overall quality and permeability of the fragment as a standalone movement area.

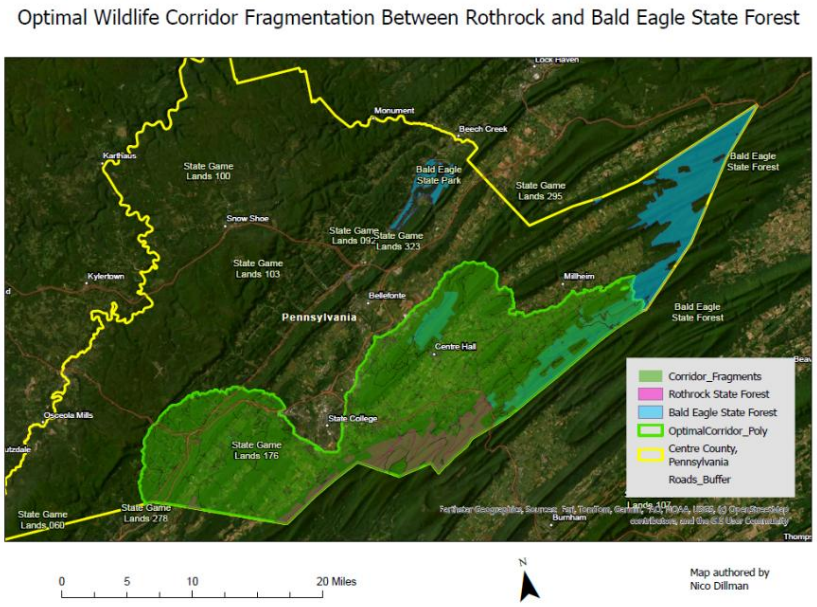


Figure 11: Overview of optimal wildlife corridor fragments (dark green) between Rothrock and Bald Eagle State Forests. The gap through the State College urban area represents the primary pinch point severing corridor connectivity.

The eastern fragment extends from the Tusseyville and Centre Hall area toward Bald Eagle State Forest, maintaining stronger connectivity with the eastern core habitat area (Figure 12). This fragment benefits from its proximity to Bald Eagle State Forest and the associated network of forested ridgelines and state game lands in the eastern portion of Centre County. The satellite imagery basemap visible beneath the fragment clearly shows the forested and agricultural character of the landscape in this area, with darker forested areas along the ridgelines contrasting with lighter agricultural fields in the valley bottoms. The eastern fragment therefore represents a more ecologically intact and functionally connected movement area compared to the western fragment, though it remains physically severed from the western fragment by the road network concentrated around State College. At this zoomed scale, Route 45 is clearly visible as a significant secondary barrier running east to west across the entire southern portion of Centre County, stretching from Musser in the west to Woodward in the east. This road runs along the southern boundary of both the western and eastern corridor fragments, effectively forming a continuous linear barrier that blocks wildlife from moving directly between the Nittany Valley lowlands and both Rothrock and Bald Eagle State Forests along the southern edge of the county. Together with Interstate 99 and Route 322, Route 45 forms part of a broader road network that collectively fragments the optimal corridor into isolated patches with limited opportunities for safe wildlife crossing.

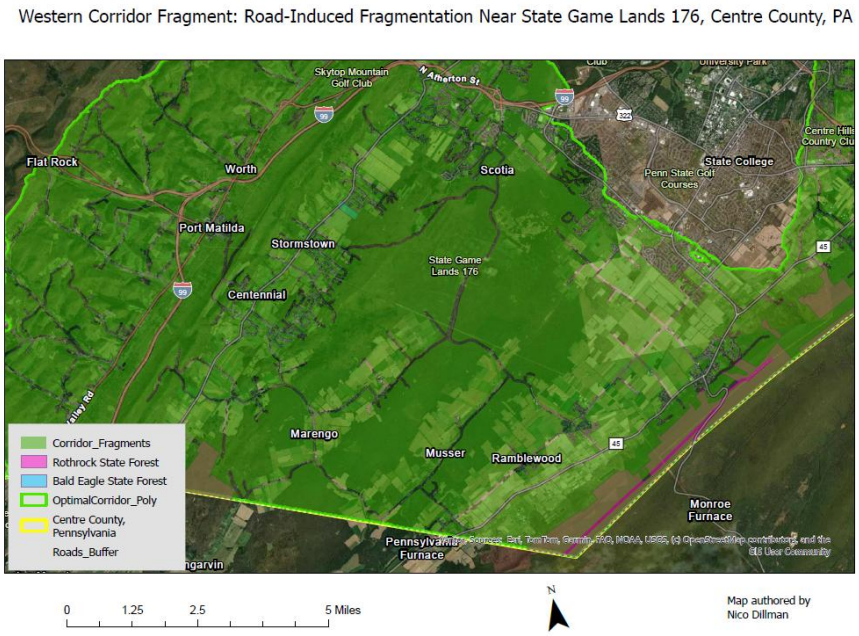


Figure 12: Close-up of the western corridor fragment near State Game Lands 176, showing road barriers along I-99 and Route 322 limiting wildlife movement.

The gap between the two fragments, centered on the State College urban area, represents the most critical pinch point where road infrastructure most severely constrains wildlife movement between Rothrock and Bald Eagle State Forests (Figure 13). The satellite imagery in this area clearly shows the contrast between the green fragmented corridor patches and the grey developed urban landscape of State College, visually reinforcing the degree to which urban development and road infrastructure have physically severed the optimal corridor. Interstate 99 and Route 322 emerge as the specific road features most responsible for this fragmentation, as they form continuous linear barriers that bisect the corridor zone without any known wildlife crossing infrastructure. Addressing this pinch point through targeted conservation interventions such as wildlife underpasses or vegetated road crossings along these two road corridors would represent the highest priority action for restoring functional connectivity between Rothrock and Bald Eagle State Forests.

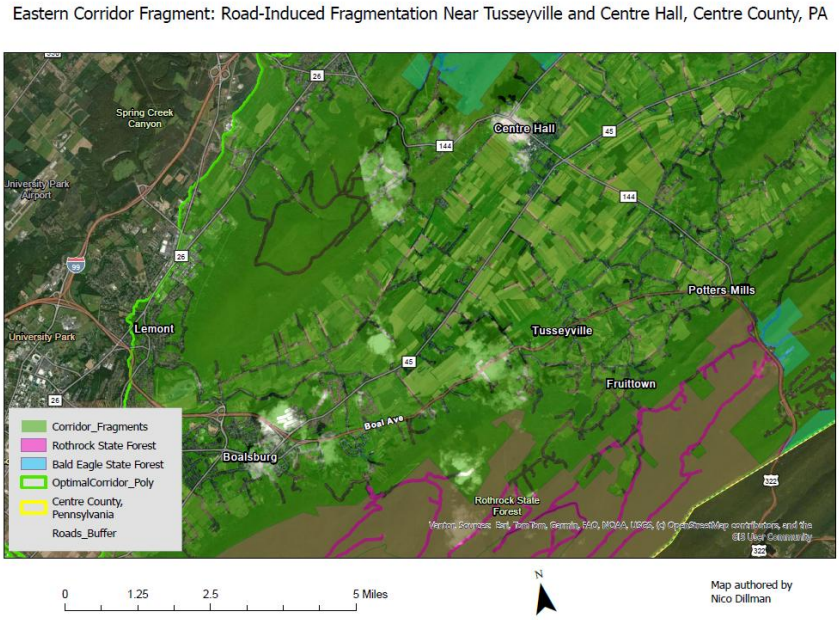


Figure 13: Close-up of the eastern corridor fragment near Tusseyville and Centre Hall, showing road-induced fragmentation along the approach to Bald Eagle State Forest.

Discussion and Reflection

This project successfully achieved its primary objectives of developing a resistance surface, modeling least-cost wildlife corridors between Rothrock and Bald Eagle State Forests and comparing corridor outputs under two different scenarios. The results directly address the research question posed at the outset of the study: How do road networks and urban development influence the location, width, and continuity of least-cost wildlife corridors between Rothrock State Forest and Bald Eagle State Forest in Centre County, Pennsylvania? The analysis demonstrates that while road networks do not fundamentally redirect the overall corridor pathway, they meaningfully constrain corridor width and suitability in developed areas, particularly around the State College urban core, and physically sever the optimal corridor into two isolated fragments. However, the process revealed several challenges and limitations worth reflecting on for future work.

The data acquisition phase presented the most significant unexpected challenges. None of the primary dataset downloads performed as expected in ArcGIS Pro, requiring all layers to be sourced through the Esri Living Atlas as a workaround. In future work, testing data sources in the analytical environment before committing to a workflow would save significant time. The GPWv4 population density layer was ultimately excluded from the analysis entirely due to its coarse 1km resolution generating widespread NoData gaps when combined with the 30-meter resolution datasets. Future work could address this using a higher resolution alternative such as LandScan USA at 90-meter resolution. The exclusion of population density as a resistance variable is a notable limitation of the present study, as human population density is widely recognized as a meaningful proxy for the intensity of human disturbance across a landscape. Areas of higher population density tend to generate greater levels of noise, light, and human activity that can deter wildlife movement even in the absence of physical barriers such as roads or buildings. The incorporation of a higher resolution population density layer in future work could therefore meaningfully alter the resistance surface and corridor outputs, particularly in the suburban fringe areas surrounding State College where population density transitions rapidly from urban to rural conditions.

The road network also presented integration challenges. The Raster Calculator provided a workable alternative to the Weighted Overlay tool, but adding a fixed resistance value of 3 to road pixels is a simplification that does not account for road type, traffic volume, or crossing difficulty. Future analyses could develop a more nuanced road resistance scheme using traffic count data or species-specific crossing probability models. The current approach treats a local residential street and Interstate 99 as equivalent road barriers once they are assigned to the same road class, which does not accurately reflect the dramatically different barrier effects these road types present to wildlife. A more refined road resistance scheme that incorporates Annual Average Daily Traffic counts from PennDOT would allow future analyses to weight road barriers in proportion to their actual traffic volume, producing a more ecologically realistic resistance surface and more accurately identifying the most critical road crossing locations for wildlife.

The weighting scheme applied to combine the resistance layers in the Raster Calculator formula also represents a source of uncertainty in the analysis. The weights assigned to land cover (45%), PAD-US (25%), and slope (30%) were based on general ecological reasoning rather than empirical data from species-specific habitat studies. Different weighting schemes could produce meaningfully different corridor outputs, as demonstrated by Parks et al. (2013), who found that varying the relative importance of landscape variables can substantially alter identified corridor pathways. A sensitivity analysis testing multiple alternative weighting schemes would be a valuable addition to future work to better characterize the uncertainty in corridor predictions associated with this methodological choice.

Working through the full analytical workflow from data preparation to final map production also gave me a much deeper appreciation for the importance of data quality and methodological transparency in GIS-based research. Early in the project, seemingly minor issues such as coordinate system mismatches and NoData conflicts between layers produced outputs that were difficult to interpret and, in some cases,

completely unusable. Resolving these issues required a significant investment of time and troubleshooting effort that was not anticipated in the original project timeline. These experiences reinforced the importance of building data validation and quality checking steps into the early stages of any GIS workflow, rather than assuming datasets sourced from reputable repositories will be immediately ready for analysis. The experience also highlighted the value of flexible tools like the Raster Calculator, which allowed me to work around the limitations of more structured tools like the Weighted Overlay by writing custom conditional expressions tailored to the specific characteristics of my data.

Despite these limitations, the resistance surface produced a spatially coherent representation of landscape permeability, the corridor modeling identified plausible movement pathways along forested ridgelines, and the fragmentation analysis physically delineated the two isolated corridor fragments severed by Interstate 99, Route 322, and Route 45. The identification of the State College urban area as the primary pinch point is a concrete and actionable finding for conservation planning. The fragmentation analysis represents a meaningful methodological contribution beyond standard least-cost corridor modeling, providing a physically explicit delineation of habitat patches separated by road barriers that could directly inform the placement of wildlife crossing infrastructure.

The findings of this project carry direct implications for conservation planning and land use decision making in Centre County. The spatial evidence produced through the resistance surface modeling, corridor analysis, and fragmentation mapping points clearly to several priority actions that could meaningfully improve wildlife connectivity between Rothrock and Bald Eagle State Forests. The most urgent conservation priority identified by this analysis is the installation of wildlife crossing infrastructure along Interstate 99 and Route 322 in the vicinity of State College, where these roads directly cut through the optimal corridor and create the primary barrier to wildlife movement between the two state forests. Wildlife underpasses or overpasses at strategic locations along these corridors, combined with exclusion fencing to guide animals toward crossing structures, represent the most direct and proven intervention for restoring functional connectivity across high-traffic road barriers. Route 45, which runs east to west across the entire southern portion of Centre County from Musser to Woodward, represents a secondary but significant priority for crossing infrastructure given its role as a continuous linear barrier along the southern boundary of both corridor fragments.

Beyond road crossings, the analysis highlights the importance of protecting the forested ridgeline corridor along Bald Eagle Mountain and the associated state game lands in the northern portion of Centre County from future development. These areas currently provide the most viable movement pathway between the two state forests and show consistently high corridor suitability across both modeling scenarios. Any future development, road expansion, or land use change in this northern corridor zone would directly threaten the primary remaining connectivity pathway between Rothrock and Bald Eagle State Forests and should be carefully evaluated for its potential impacts on wildlife movement. Land conservation organizations and state agencies operating in Centre County should prioritize the acquisition or permanent protection of privately owned parcels within and adjacent to this northern corridor zone to ensure its long-term availability for wildlife movement.

Finally, the State College metropolitan area itself represents a long-term land use planning challenge for wildlife connectivity. While the construction of wildlife crossings can address road barriers at specific locations, the broader pattern of urban and suburban development spreading outward from State College into the surrounding agricultural and semi-natural lands of the Nittany Valley represents a slower but equally significant threat to corridor connectivity. Incorporating wildlife corridor considerations into local and regional land use planning processes, including zoning decisions, development approvals, and transportation planning, would help ensure that future growth in the State College area does not further erode the limited movement opportunities that currently exist for wildlife attempting to cross between the two state forests. The spatial evidence produced by this project provides a direct and accessible

foundation for those planning conversations, clearly showing where connectivity is most critical and where development pressure poses the greatest risk to long term landscape permeability.

Extensions of this work could include species-specific corridor modeling targeting wildlife commonly found in Centre County such as white-tailed deer, black bear, wild turkey, or bobcat. Future work could also validate the modeled corridors by comparing them against actual wildlife sighting records or tracking data to confirm whether animals are truly moving through the identified pathways. Additionally, incorporating wildlife occurrence data from citizen science platforms such as iNaturalist could provide an accessible and visually compelling layer of biological evidence to support the modeled findings. A temporal analysis comparing NLCD land cover data across 2001, 2011, and 2021 could further reveal how habitat connectivity in Centre County has changed over time in response to land use change and the continued expansion of development in the Nittany Valley.

Building on this, a logical next step would be to move from identifying general corridor zones to pinpointing exact locations for wildlife crossing implementation. This could be done by intersecting the optimal corridor with major road networks to identify where high-quality movement areas are directly constrained by infrastructure. Locations where strong corridor suitability overlaps with high-traffic roads would represent priority sites for wildlife crossings or mitigation efforts, as they would provide the greatest benefit for both animal movement and driver safety. Interstate 99, Route 322, and Route 45 represent the most immediate candidates for wildlife crossing assessment given their demonstrated role in severing the optimal corridor identified in this study.

Overall, this project shows how GIS-based corridor modeling can identify likely wildlife movement pathways and clearly highlight where connectivity breaks down, particularly around high-impact areas like State College. The combination of resistance surface modeling, least-cost corridor analysis, scenario comparison, and corridor fragmentation analysis provided a comprehensive and spatially explicit picture of wildlife movement opportunities and constraints in Centre County, demonstrating the power of GIS as a tool for evidence-based conservation planning.

References

Beltrão, M. G., et al. (2024). Priority areas and implementation of ecological corridors. Scientific Reports.

Esri. (2024). ArcGIS Pro. Environmental Systems Research Institute.

Esri. (2024). PAD-US Protection Status by GAP Status Code. Living Atlas of the World. ArcGIS Online.

Esri. (2024). USA Annual NLCD Land Cover. Living Atlas of the World. ArcGIS Online.

Esri. (2024). USGS 3DEP Elevation 30m. Living Atlas of the World. ArcGIS Online.

Esri. (2024). World Imagery Hybrid. Living Atlas of the World. ArcGIS Online.

Liu, C., et al. (2018). Identifying wildlife corridors for the restoration of regional habitat connectivity. PLOS ONE.

NASA SEDAC. (2020). Gridded Population of the World, Version 4 (GPWv4): Population Density. Center for International Earth Science Information Network.

OpenStreetMap contributors. (2024). OpenStreetMap. OpenStreetMap Foundation. https://www.openstreetmap.org

Parks, S. A., et al. (2013). Effects of weighting schemes on wildlife corridors. Conservation Biology.

Pennsylvania Department of Conservation and Natural Resources. (2024). Bald Eagle State Forest. Bureau of Forestry. https://www.dcnr.pa.gov/StateForests/FindAForest/BaldEagle/Pages/default.aspx

Pennsylvania Department of Conservation and Natural Resources. (2024). Rothrock State Forest. Bureau of Forestry. https://www.dcnr.pa.gov/StateForests/FindAForest/Rothrock/Pages/default.aspx

R Core Team. (2024). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing. https://www.r-project.org/

U.S. Census Bureau. (2025). TIGER/Line Shapefiles: Roads, Centre County, Pennsylvania.

USGS. (2024). Protected Areas Database of the United States (PAD-US) 4.1. U.S. Geological Survey Gap Analysis Project.

Wickham, H. (2016). ggplot2: Elegant Graphics for Data Analysis. Springer-Verlag.

Xu, J., et al. (2024). Effects of habitat fragmentation on habitat quality. Ecological Indicators.